Chapter 21 Review

Campbell Biology, AP Edition, 12e

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Summary of Key Concepts

Scientists have sequenced many model organisms and estimated the number of protein coding genes in each, including rice, 36,000; the nematode *C. elegans*, 20,000; and humans, 21,300. Explain the significance of these findings. **(BIG IDEAS 1 & 3)**

Concept 21.1: The Human Genome Project fostered development of faster, less expensive sequencing techniques

- ****The ****Human Genome Project** was largely completed in 2003, aided by major advances in sequencing technology.
- ****In the ****whole-genome shotgun** approach, the whole genome is cut into many small, overlapping fragments that are sequenced; computer software then assembles the genome sequence.

How did the Human Genome Project result in more rapid, less expensive DNA-sequencing technology?

Concept 21.2: Scientists use bioinformatics to analyze genomes

and their functions

- ****Computer analysis of genome sequences aids ****gene annotation**, the identification of protein-coding sequences. Methods to determine gene function include comparing sequences of newly discovered genes with those of known genes in other species and observing the effects of experimentally inactivating the genes.
- ****In ****systems biology**, scientists use the computer-based tools of **bioinformatics** to compare genomes and study sets of genes and proteins as whole systems (****genomics** and ****proteomics**). Studies include large-scale analyses of protein interactions, functional DNA elements, and genes contributing to medical conditions.

What has been the most significant finding of the ENCODE project? Why was the project expanded to include nonhuman species?

Concept 21.3: Genomes vary in size, number of genes, and gene

density

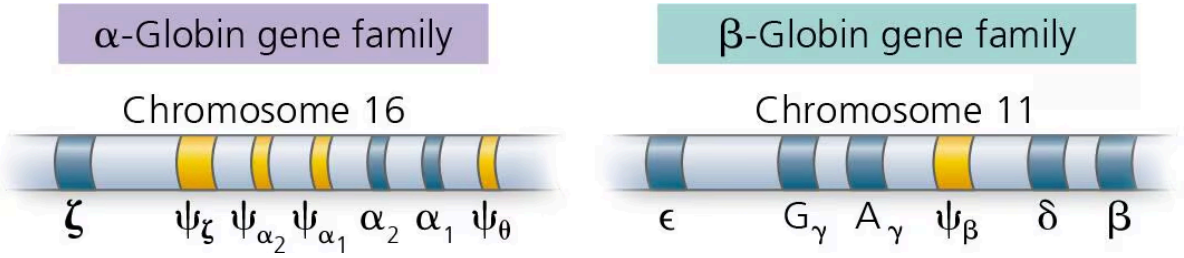
	Bacteria	Archaea	Eukarya
Genome size	Most are 1–6 Mb		Most are 10–4,000 Mb, but a few are much larger
Number of genes	1,500–7,500		Most are 5,000–45,000
Gene density	Higher than in eukaryotes		Lower than in prokaryotes (Within eukaryotes, lower density is correlated with larger genomes.)
Introns	None in protein-coding genes	Present in some genes	Present in most genes of multicellular eukaryotes, but only in some genes of unicellular eukaryotes
Other noncoding DNA	Very little		Can exist in large amounts; generally more repetitive noncoding DNA in multicellular eukaryotes

blank	Bacteria	Archaea	Eukarya
Genome size	Most are 1 to 6 megabytes	Most are 1 to 6 megabytes	Most are 10 to 4000 megabytes
Number of genes	1500 to 7500	1500 to 7500	Most are 5000 to 45,000
Gene diversity	Higher than in eukaryotes	Higher than in eukaryotes	Lower than in prokaryotes (Within eukaryotes, lower density is correlated with large genomes.)
Introns	None in protein-coding genes	Present in some genes	Present in most genes of multicellular eukaryotes, but only in some genes of unicellular eukaryotes
Other noncoding D N A	Very little	Very little	Can exist in large amounts; generally more repetitive noncoding D N A in multicellular eukaryotes

Compare genome size, gene number, and gene density (a) in the three domains and (b) among eukaryotes.

Concept 21.4: Multicellular eukaryotes have a lot of noncoding DNA and many multigene families

- ****Only 1.5% of the human genome codes for proteins or gives rise to rRNAs or tRNAs; the rest is noncoding DNA, including ****__pseudogenes__ and ****__repetitive DNA__ of unknown function.
- The most abundant type of repetitive DNA in multicellular eukaryotes consists of ****__transposable elements__ and related sequences. In eukaryotes, there are two types of transposable elements: ****__transposons__, which move via a DNA intermediate, and ****__retrotransposons__, which are more prevalent and move via an RNA intermediate.
- Other repetitive DNA includes short, noncoding sequences that are tandemly repeated thousands of times (****__simple sequence DNA__, which includes ****__STRs__); these sequences are especially prominent in centromeres and telomeres, where they probably play structural roles in the chromosome.
- Though many eukaryotic genes are present in one copy per haploid chromosome set, others (most, in some species) are members of a gene family, such as the human globin gene families:



Key: Light blue = noncoding regions. Dark blue = gene. Gold = pseudogene.

Alpha-globin gene family. Chromosome 16 regions, left to right: Noncoding region; zeta gene with arrow to embryo; noncoding region; psi zeta gene; noncoding region; psi alpha 2 pseudogene; noncoding region; psi alpha 1 pseudogene; noncoding region; alpha 2 gene with arrow to fetus and adult; noncoding region; alpha 1 gene with arrow to fetus and adult; noncoding region; psi theta pseudogene; noncoding region.

Beta-globin gene family. Chromosome 11 regions, left to right: Noncoding region; epsilon gene with arrow to embryo; noncoding region; gamma G gene with arrow to fetus; noncoding region; gamma A gene with arrow to fetus; noncoding region; psi beta pseudogene; noncoding region; delta gene with arrow to adult; noncoding region; beta gene with arrow to adult; noncoding region.

Explain how the function of transposable elements might account for their prevalence in human noncoding DNA.

Concept 21.5: Duplication, rearrangement, and mutation of DNA contribute to genome evolution

- Errors in cell division can lead to extra copies of all or part of entire chromosome sets, which may then diverge if one set accumulates sequence changes. Polyploidy occurs more often among plants than animals and contributes to speciation.
- The chromosomal organization of genomes can be compared among species, providing information about evolutionary relationships. Within a given species, rearrangements of chromosomes are thought to contribute to the emergence of new species.
- The genes encoding the various related but different globin proteins evolved from one common ancestral globin gene, which duplicated and diverged into the α-globin and β-globin ancestral genes. Subsequent duplication and random mutation gave rise to the present globin genes, all of which code for oxygen-binding proteins. The copies of some duplicated genes have diverged so much that the functions of their encoded proteins (such as lysozyme and α-lactalbumin) are now substantially different.
- Rearrangement of exons within and between genes during evolution has led to genes containing multiple copies of similar exons and/or several different exons derived from other genes.
- Movement of transposable elements or recombination between copies of the same element can generate new sequence combinations that are beneficial to the organism. These may alter the functions of genes or their patterns of expression and regulation.

How could chromosomal rearrangements lead to the emergence of new species?

Concept 21.6: Comparing genome sequences provides clues to evolution and development

- Comparisons of genomes from widely divergent and closely related species provide valuable information about ancient and more recent evolutionary history, respectively. Analysis of single nucleotide polymorphisms (SNPs) and copy-number variants (CNVs) among individuals in a species can also shed light on the evolution of that species.
- Evolutionary developmental (****__evo-devo)__ biologists have shown that homeotic genes and some other genes associated with animal development contain a ****__homeobox__ region whose sequence is highly conserved among diverse species.

What type of information can be obtained by comparing the genomes of closely related species? Of very distantly related species?